

**All Subjects Notes, Past Quizzes &
Past Papers**

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State Continuous Mapping Theorem ②

Define basis of finite dimensional vector space. ②

Define separable space and dense set. ③

Write canonical form of $\mathbb{R}^n$ ③

Show that the space l^∞ is complete ⑤

Given a norm space $(X, \|\cdot\|)$,

Prove that $|\|x\| - \|y\|| \leq \|x - y\|$

Q Draw the unit sphere for the norm
define by $\|x\|_2 = (|x_1|^2 + |x_2|^2)^{1/2}$ in $\mathbb{R}^2$

Q Show that the space of all the bounded
functions on a set A is a metric space
define as.

$$d(x, y) = \sup_{t \in A} |x(t) - y(t)|$$

Q Show that the identity operator $I: X \rightarrow X$
on a normed space $X \neq \{0\}$ is bounded
and has norm $\|I\| = 1$

Q Find countable subset L^p for $1 \leq p < +\infty$

Q Define Riesz's Lemma.

Q Show rational numbers $\mathbb{Q}$ are dens in $\mathbb{R}$.

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

22/6/2019

Mth 641

Current paper:-

- 1) Define the neighbourhood of a point in a metric space. (2)
- 2) Define a compact metric space. X (2)
- 3) Define convergent series in normed space (3)
- 4) Show that A subspace M is complete metric X is itself (3)
complete if the set M is closed in X .
- 5) Show that space ℓ^∞ is convergent (5)
- 6) state prove compactness theorem. (5)

MTH 641. Paper

Show that absolutely convergent series is normed space,
define complete metric space and....

define linear operator

Prove closure of M is.....

X and Y are spaces then $T:D(T) \rightarrow Y$ is linear operator if T^{-1} exists if
and only if $Tx=0$ implies that $x=0$

Prove Real line is a complete metric space

7:32 PM

➡ Forwarded

MTH 641. Paper

1): Show that C is a Separable space,

2): compact space,

3): Give two examples of linear operator

4): Prove x belong to closure of M .

5): X and Y are spaces then $T:D(T) \rightarrow Y$ is linear operator if T^{-1} exists.

Show that it is a linear operator. 6

6): Finite dimension prove

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FALL/SPRING MID/FINAL TERM 20__ (year)

Rough Sheet

Date: 3/1Superintendent Signature: [Signature]Campus Code: UICD01Student Signature: [Signature]Course Code: MTM 641Student ID: [ID]

- ② In a metric space (X, d) every convergent sequence is Cauchy Seq.
- ③ The seq $(\frac{1}{n})$ is strictly decreasing.
- ④ A seq (x_n) in a metric space is said to be strictly increasing if for all $n, x_n < x_{n+1}$.
- ⑤ Which of the following metric spaces is incomplete? Real space $\mathbb{R}$
- ⑥ Two metric spaces (X, d) & (Y, ρ) are said to be isometric if there exists an isometric isometry of X onto Y .
- ⑦ Which of the following axiom holds for any vectors u, v, w in vector space V , scalar in field K ? Ans: $(u+v)+w = u+(v+w)$
- ⑧ Let X be an n -dimensional vector space then any proper subspace Y of X has dimension equal to n .
- ⑨ A sequence (x_n) in a normed space X is convergent if it contains an $l \in X$.
- ⑩ Let T be a linear operator, and if $\dim(D(T)) = \dim(R(T)) = 0$.
- ⑪ $d(x, z) \leq d(x, y) + d(y, z) \rightarrow$ Triangle property.
- ⑫ $C[0, b]$ shows the space of all continuous functions defined on $[0, b]$.
- ⑬ $x \notin B(x, r) \rightarrow d(x, x) \geq r$.
- ⑭ A discrete metric space X is Complete iff X is countable.
- ⑮ A metric space X is said to be compact if every seq in X has a convergent subsequence.
- ⑯ Every finite dimensional normed space is Complete.
- ⑰ Which of the following is not a bounded operator? Integral operator.

3. Show that $d(x, y) = \sqrt{|x - y|}$ defines a metric on the set of all real numbers.

Solution: Fix $x, y, z \in X = \mathbb{R}$, we need to verify the axioms of a metric. (M1) to (M3) follows easily from properties of absolute value. To verify (M4), for any $x, y, z \in \mathbb{R}$ we have

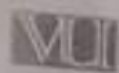
$$\begin{aligned} \left[d(x, y) \right]^2 &= |x - y| \leq |x - z| + |z - y| \\ &\leq |x - z| + |z - y| + 2\sqrt{|x - z|}\sqrt{|z - y|} \\ &= (\sqrt{|x - z|} + \sqrt{|z - y|})^2 \\ &= \left[d(x, z) + d(z, y) \right]^2. \end{aligned}$$

Taking square root on both sides yields the triangle inequality.

5. Is boundedness of a sequence in a metric space sufficient for the sequence to be Cauchy? Convergent?

Solution: It turns out that boundedness of a sequence in a metric space does not imply Cauchy or convergent. Consider the sequence (x_n) , where $x_n = (-1)^n$; it is clear that (x_n) is bounded in $(\mathbb{R}, d)$, where $d(x_m, x_n) = |x_m - x_n|$.

- (x_n) is not Cauchy in $\mathbb{R}$ since if we pick $\varepsilon = \frac{1}{2} > 0$, then for all N , there exists $m, n > N$ (we could choose $m = N + 1$, $n = N + 2$ for example) such that $d(x_m, x_n) = d(x_{N+1}, x_{N+2}) = 1 > \frac{1}{2}$.
- (x_n) is not convergent in $\mathbb{R}$ since it is not Cauchy.



SPRING MID TERM 2019

Campus Code: PH8002

Course Code: ACT541

Rough Sheet

Date: 10/11/2019

Supervisor Signature: [Signature]

Student Signature: [Signature]

Student ID: 201905000000

- ① write the usual metric " d " in the Euclidean Plane $\mathbb{R}^2$ and Euclidean space $\mathbb{R}^n$
- ② write the norm on $\mathbb{R}^n$ for which it is a Banach space
- ③ If T is a linear operator, then show that the null space $N(T)$ is a vector space
- ④ Prove that $n \in M$ if there is a surjective map $n \rightarrow M$ such that $n \rightarrow M$
- ⑤ Let X be an n -dimensional vector space. Then show that any proper subspace Y of X has dimension less than n .

Notes: Submit this sheet to Supervisor, before leaving the Examination Center.

- Q1. $B(x_0, 1) \subset V$. Justify its elements.
- Q2. Define Series in normed spaces.
- Q3. Differentiate between finite dimensional vector space and infinite dimensional vector space.
- Q4. Prove that every convergent series in a metric space is a Cauchy sequence.
- Q5. Prove $d(x, y) = \|x - y\|$ triangle inequality.
- Q6. Prove that $\|\cdot\|$ is equivalent to $\|\cdot\|_0$.

"Current Paper"

(MTH-641)

21-June-2019

- ① Definition of Accumulation point. (2)
- ② Prove that Real line is metric space. (2)
- ③ Give 4-examples of linear operator. (3)
- ④ Give 5-examples of norm space. (3)
- ⑤ Prove that $(ST)^{-1} = T^{-1}S^{-1}$ (5)
- ⑥ Theorem prove Krna tha continuous mapping ka. (5)

* MCQ's ~~come~~ (very easy)

⇓
Mostly from theorems and Lemmas' statements.

Norm space ki 5 examples

1:04 PM

D (3,2) ki vector spaces find krni
thi

1:05 PM

$ST = TS$ inverse prove krna tha

1:06 PM

Deff of continuous

1:07 PM

Continuous wala theorm

1:07 PM

Define convergence in normed space. Differentiate between finite dimensional vector space and

12:13 pm

And infinite dimensional space, prove that every sequence convergence in a metric space is a Cauchy sequence.

12:13 pm

Mth641 pppr 12:14 pm



Mth641 my paper

Define series of normed space

Give 5 examples of normed space

Is boundedness of a sequence in a metric space sufficient for the sequence to be Cauchy? or convergent?

Show that $d(x,y) = |x-y|$ defines a metric on the set of all real numbers

mth641 today's paper
triangle inequality prove krni
thi.(5)
bounded linear operator
define.(3)
cauchy sequency ki
convergent btani thi $x_n = 1 + 1/n$
ki(2)
linear operator ki 4 examples
(2)
compactness ka theorem tha.
(5) mujy nai ata tha sai wala
isometric ki defination (3)